# Estimation of Multi-state Dependent Disturbance by Using Multi-dimensional Gaussian Process

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***Abstract*—High-precision linear motor stages have been widely used for their excellent positioning accuracy and speed. However, core-type linear motor stages have performance limitations because of various nonlinear factors including cogging force, friction, and geometrical imbalance. This paper analyzes disturbances in velocity and position domains and trains a Two-Input-Single-Output (TISO) nonlinear model using the Gaussian process for the disturbance. With this, two state-dependent disturbances are removed effectively. As a result, the control performance with a proposed controller is enhanced. Ultimately, this paper introduces three contribution points: 1) analysis of disturbances based on position/velocity, 2) design of TISO Gaussian process model, and 3) validation of estimation performance of proposed algorithm through simulation.**

## I. INTRODUCTION

In the industrial field, a linear motor has been achieved high control bandwidth with easy maintenance, high accuracy, and fast speed [1]. However, it is sensitive to external disturbances such as force ripple and friction [2]. Numerous studies for linear motors aim to eliminate such unpredictable nonlinear disturbances and enhance control performance.

Especially, disturbances occurring in core-type linear motors can be identified based on the geometry information of fixed magnets [3]. In many studies, disturbances in the linear motor have been modeled using harmonic polynomial or multi-sinusoidal functions [4]. However, there is a challenge in adjusting the level of overfitting, which has reduced the reliability of the models. Additionally, the Gaussian process method, beneficial for modeling nonlinearity, has been utilized. While this method satisfactorily models nonlinear disturbances, it describes the relationship from a single-input-single-output (SISO) perspective [5]. Since [6] verified that disturbances in linear actuators affect position and velocity domains, there are limitations to modeling disturbances in linear motors as SISO model.

The Gaussian process (GP) method has been applied to various control systems by learning nonlinearity based on data. Particularly, many studies have been published by combining GP and robust model-based controllers. [7] is designed with computed torque control to eliminate unexpected disturbances, while GP is utilized to compensate torque ripple with the repetitive controller by modeling disturbances at arbitrary velocity [5]. [8] proposed GP-based compensation with covariance gain to remove cogging force in linear actuators with disturbance observer(DOB). However, [5], [7], [8] utilize GP to create SISO disturbance models. However, the issue focused on in this study is the two state dependent disturbances, which depends simultaneously on position and velocity. Therefore, it is necessary to remove two state-dependent disturbances and improve control performance using an expanded-dimensional GP.

In this paper, the proposed algorithm using a two-dimensional GP model is developed to suppress disturbances described by two states of dynamic system. Core-type linear motors suffer from unspecified nonlinear factors, including cogging force, well-known as a function of position. However, velocity-dependent disturbances practically affects the control performance, as revealed in various studies. To overcome this, a two-dimensional Gaussian process is employed to learn two state-dependent disturbances. The learned model is utilized with a robust controller, specifically a DOB, to compensate for disturbances. Furthermore, the proposed algorithm's performance is assessed by establishing a practical simulation environment, followed by comparing its control efficacy with that of existing methods.

In summary, the main contributions of this paper are as follows:

1) The characteristics of disturbances in linear motor dependent on position and velocity are experimentally verified and modeled.
2) A Two-dimensional Gaussian process model is designed for disturbances in the position and velocity domains.
3) A simulation setup is established to verify the effectiveness of the proposed algorithm estimating disturbances related to both position and velocity.

## II. PROBLEM STATEMENT

### A. Modeling Force Ripple in Linear Motor

Disturbances generated in linear motor with powerful thrust is mainly related to the interaction between the iron core of the mover and the magnets of the stator. In iron core-type linear motors, depicted in Fig. 1, which move linearly, the mover is affect by the undefined forces due to the strong magnets uniformly distributed on the stator. The neodymium magnets on the stator are arranged at geometrically constant

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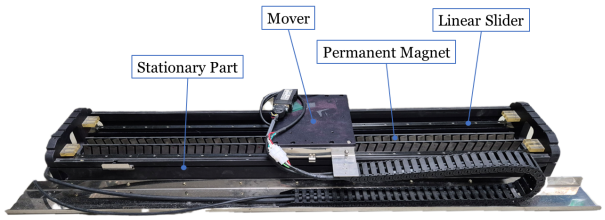


Figure 1. Linear motor stage system

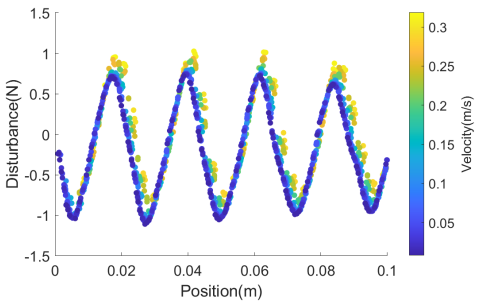


Figure 2. Position and velocity varying disturbances of linear motor.

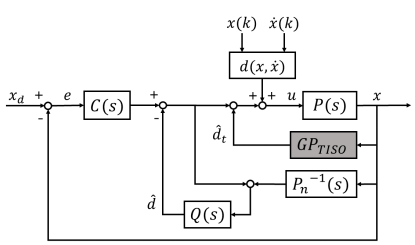


Figure 3. Block diagram of the proposed controller including TISO GP model of disturbances.

intervals, causing the rotor to be influenced by the magnetic field periodically depending on its position.

Fig.2 shows the disturbance information obtained when operating the linear motor at different velocities within the specific position range. This information is obtained when the control structure without $GP_{TISO}$ is appropriately tuned as depicted in Fig.3. The estimated disturbances $\hat{d}$ are obtained using the DOB structure and used to train $GP_{TISO}$. Because of the causality problem of DOB, a Q-filter, $Q(s)$, is added. Through Fig.2, it is evident that practical disturbances in the linear motor vary not only with position but also with velocity. Consequently, it can be considered that actuators operating in structures with magnets positioned at equidistant intervals are influenced by position and velocity-dependent magnetic forces.

### B. Expansion of GP model from 1D to 2D

In the previous study [8], disturbances are modeled solely based on position using GP. Therefore, considering that disturbances are entirely a function of position and that unmodeled disturbances are appropriately rejected by the existing DOB, a one-dimensional GP modeling for periodic disturbances solely dependent on position posed no issues. However, the actual disturbances in the linear motor, as depicted in Fig. 2, are not perfectly periodic with respect to position, and the amplitude also varies slightly. Additionally, since bias occurs with velocity, it is not appropriate to model and compensate for disturbances based solely on information related to position.

Hence, to compensate for modeling errors arising from insufficient information, it is necessary to extend the dimension of the GP model by considering two types of information: position and velocity. Among various factors influencing uncertainty of disturbance, it can be observed from Fig. 2 that position and velocity predominantly influence disturbances.

However, unlike 1-Dimensional (1D) GP, 2-Dimensional (2D) GP requires adequate information considering both position and velocity. The 1D GP model is a single-input-single-output GP (SISO GP) that models disturbances based solely on position, while the 2D GP model is a two-input-single-output GP (TISO GP) that models disturbances based on both position and velocity.

## III. MULTI-DIMENSIONAL GAUSSIAN PROCESS

Generally, GP is well-known for modeling functions over distributions and is well-recognized as one of the non-parametric stochastic modeling techniques for nonlinear modeling. Compared to deterministic parametric modeling methods, algorithms utilizing GP are relatively simple but are influenced by the speed and performance of modeling depending on the number of training data points. Especially, if the quality of the training data is good, meaning it accurately represents the characteristics of the function, it is possible to sufficiently model the function even with a small amount of data.

### A. GP Model for Multi-Dimensional Disturbance

In this study, disturbance considering both position and velocity is modeled using GP, and it is utilized in the proposed algorithm shown in Fig. 3 to compensate for the disturbance. The modeled disturbance($d(X)$) is dependent on both position and velocity, and observation($d_n$) is obtained in practical environments with measurement noise which is considered by GP. The state vector $X = \begin{bmatrix} x & \dot{x} \end{bmatrix}^\top \in \mathbb{R}^{2\times 1}$ determining the disturbance includes position ($x$) and velocity ($\dot{x}$). Therefore, the observation is as follows,

$$d_n(X) = d(X) + \epsilon, \text{ with } \epsilon \sim \mathcal{N}(0, \sigma_n^2), \tag{1}$$

where $d(X)$ is the disturbance obtained by DOB structure, $\epsilon$ is the measurement noise, which is independent identically distributed zero-mean Gaussian noise with variance $\sigma_n^2$.

The set of $N$ states and disturbances used for disturbance estimation is defined as follows,

$$\boldsymbol{X} = \begin{bmatrix} X_1 & X_2 & \cdots & X_N \end{bmatrix}^\top, \tag{2a}$$

$$\boldsymbol{d}_n = \begin{bmatrix} d_{n,1} & d_{n,2} & \cdots & d_{n,N} \end{bmatrix}^\top, \tag{2b}$$

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The joint distribution of the new disturbance data ($\tilde{\boldsymbol{d}}_*$) for $M$ test states $\boldsymbol{X}_* \in \mathbb{R}^M$, including state and disturbance observations $\mathcal{D}(\boldsymbol{X}, \boldsymbol{d}_n)$, is as follows,

$$\begin{bmatrix} \boldsymbol{d}_n \\ \tilde{\boldsymbol{d}}_* \end{bmatrix} \sim \mathcal{N}\left( \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} K + \sigma_n^2 I_N & K_* \\ K_*^\top & K_{**} \end{bmatrix} \right), \tag{3}$$

where $K \in \mathbb{R}^{N \times N}$ is the kernel matrix of train data $k(\boldsymbol{X}, \boldsymbol{X})$, and similar for $K_* = k(\boldsymbol{X}, \boldsymbol{X}_*) \in \mathbb{R}^{N \times M}$ and $K_{**} = k(\boldsymbol{X}_*, \boldsymbol{X}_*) \in \mathbb{R}^{M \times M}$ are kernel matrices, with $N$ the number of observations and $M$ the number of test points.

The posterior distribution of $\tilde{\boldsymbol{d}}_*$ for the test data $\boldsymbol{X}_*$ is given by,

$$p(\tilde{\boldsymbol{d}}_* | \mathcal{D}, \boldsymbol{X}_*) = \mathcal{N}(\mu_d, \Sigma_d), \tag{4}$$

where $\mu_d$ and $\Sigma_d$ are the posterior mean and covariance [9], respectively, as follows,

$$\mu_d = K_*^\top (K + \sigma_n^2 I_N)^{-1} \boldsymbol{d}_n, \tag{5a}$$

$$\Sigma_d = K_{**} - K_*^\top (K + \sigma_n^2 I_N)^{-1} K_*, \tag{5b}$$

where $I_N \in \mathbb{R}^{N \times N}$ is an identity matrix. The unobserved disturbance affecting to the system can be estimated by (5a).

The proposed controller ensures high control performance by appropriately utilizing (5a). For the real-time implementation of GP control, the estimated disturbance $\hat{d}_*$ for a single test data $X_*$ is compensated at each sampling instant by computing it as the posterior mean. However, the process of obtaining the inverse matrix in (5a) requires considerable computational resources. Therefore, to alleviate this, after obtaining the train data, $\alpha$ is precomputed for GP prediction and hold it as follows,

$$\alpha = (K + \sigma_n^2 I_N)^{-1} \boldsymbol{d}_n, \tag{6a}$$

### B. Multi-Dimensional Kernel for GP

The selection of an appropriate kernel function can significantly affect the efficiency and performance of GP regression. In this paper, to train a TISO GP model for the disturbance $d_n \sim \mathcal{GP}(0, k(X_p, X_q))$, we will utilize the square exponential kernel to form a smooth model by referencing neighboring data. $k(X_p, X_q)$ is the kernel function between one state ($X_p$) and another state ($X_q$), following the expression,

$$k(X_p, X_q) = \sigma_f^2 \exp(-\frac{1}{2} \bar{X}^\top L^{-1} \bar{X}) \tag{7}$$

where $\bar{X} = (X_p - X_q)$ and $L = \text{diag}(\ell_1^2, \ell_2^2)$. Unlike the 1D kernel, there are two length scales ($\ell_1, \ell_2$), which means setting reference lengths for each dimension of the data.

## IV. SIMULATION CASE STUDY

### A. Simulation Environment Setting

The simulation configuration is depicted in Fig. 3. $P(s)$ is set as a second-order mass-damper system,

$$P(s) = \frac{1}{Js^2 + Bs} \tag{8}$$

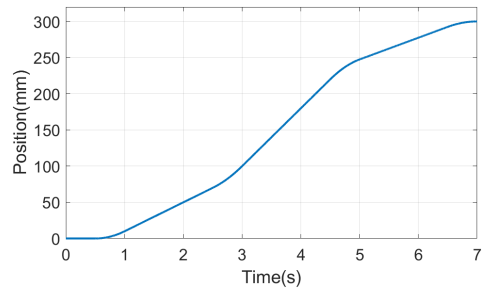


Figure 4. Position reference used in the simulation

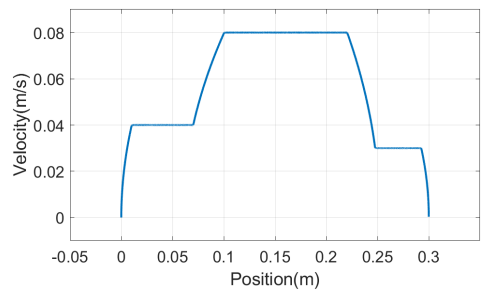


Figure 5. The region of position and velocity reference

where inertia $J = 0.238 \cdot 10^{-3}\text{kg} \cdot \text{m}^2$, damping $B = 3.810 \cdot 10^{-3}\text{kg} \cdot \text{m}^2/\text{s}$, and $C(s)$ which is designed as given by,

$$C(s) = 22.5591 + 472.4766\frac{1}{s} + 0.1496s \tag{9}$$

The gains of PID controller are manually tuned to obatin disturbance information for specific positions and velocities.

The disturbance used in the simulation is described by,

$$d(x, \dot{x}) = A \sin(\frac{2\pi}{x_p}(x - \phi_a \dot{x})) + \phi_b \dot{x} \tag{10}$$

where $A = 0.9$, $x_p = 0.022$m, $\phi_a = 0.05$, and $\phi_b = 0.9$ which effectively characterizes disturbances of the linear motor as shown in Fig. 2.

### B. The Region of Training Dataset

To conduct the simulation, it is necessary to identify the motion range of the system and then obtain data to apply to train the GP model. The operating range of the system for the reference in Fig. 4 corresponds to Fig. 5. The range of positions is 0∼0.3m, and the range of velocities is 0∼0.08m/s. For TISO GP regression within this range, the data acquisition range is set to positions 0∼0.4m and velocities 0∼0.1m/s. To obtain data within this range, position control is conducted at constant speeds, which are 0.01, 0.05, and 0.1 m/s, as shown in Fig. 6.

### C. Hyperparameter setting of 2D GP kernel

**Example 1. 1D kernel induced disturbance model**

To compare the effectiveness, 1D kernel case is set to obtain the results. Since its performance is shown based on 1D data, the state $X$ includes only the position information

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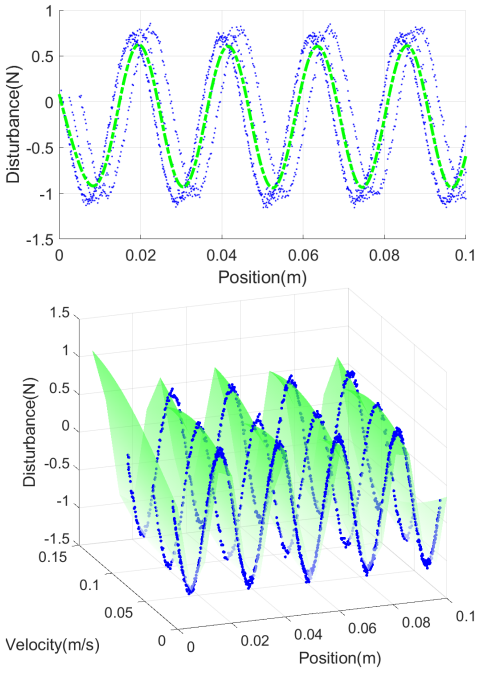


Figure 6. Regression results of GP trained with 1D (Top) and 2D (Bottom) kernels. Actual disturbances based on each dimension of the state (—), and regression results of SISO(- -), TISO(■) GP trained based on actual disturbances.

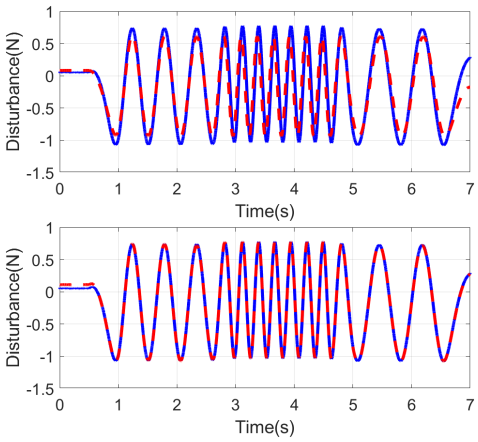


Figure 7. Estimation of disturbance by SISO (Top) and TISO (Bottom) GP in time domain. Actual disturbances (—), estimated disturbances by GP (- -).

$x$. Therefore, the hyperparameters, in this case, are defined as $\ell_{1\mathrm{D}}$ and $\sigma_{f,1\mathrm{D}}$, as follows,

$$\theta_{1\mathrm{D}} = \begin{bmatrix} \ell_{1\mathrm{D}} & \sigma_{f,1\mathrm{D}} \end{bmatrix} \tag{11}$$

The optimal hyperparameters can be obtained by maximizing the marginal log likelihood [9]. The results are as follows: $\ell_{1\mathrm{D}} = 0.0095$ and $\sigma_{f,1\mathrm{D}} = 1.5417$. Using the 1D kernel defined with these hyperparameters, the SISO GP regression results are shown in the top plot of Fig. 6. Since the magnitude of disturbances varies with the velocity for a given position, it cannot accurately represent all observations.

**Example 2. 2D kernel induced disturbance model**

The TISO GP regression method proposed in this paper considers both position and velocity to learn the disturbance model compared to SISO GP. Thus, the optimal hyperparameters are $\ell_1 = 0.0058$, $\ell_2 = 0.148$, and $\sigma_f = 2.4633$. The TISO GP regression results are shown in the bottom plot of Fig. 6. It can cover variations in disturbances with respect to velocity. In the bottom plot of Fig. 6, although there is no data in the velocity range of $0.05{\sim}0.1$m/s, the TISO GP estimates for disturbances data which can be obtained using the posterior mean based on nearby data.

The results of estimating disturbances using the GP model are shown in Fig. 7. The estimation performance of the SISO GP is poor in the region of high velocity ($3 \sim 4.5$s), which is attributed to the model being trained without considering velocity. In particular, since velocity is not considered, the estimation performance in the peak regions of the disturbances is varying as velocity changes.

## V. CONCLUSION AND FUTURE WORK

In this paper, the proposed method estimates the disturbances influenced by two states. Therefore, the TISO GP model is designed to estimate the two-state dependent disturbances and apply the GP model to the controller. The proposed method effectively estimated disturbances varying in position and velocity, which is verified in a practical simulation environment compared to the conventional methods.

## REFERENCES


[1] R. Hecker, G. Flores, Q. Xie, and R. Haran, "Servocontrol of machine-tools: a review," *Latin American applied research*, vol. 38, no. 1, pp. 85–94, 2008.

[2] F. J. Villegas, R. L. Hecker, M. E. Peña, D. A. Vicente, and G. M. Flores, "Modeling of a linear motor feed drive including pre-rolling friction and aperiodic cogging and ripple," *The International Journal of Advanced Manufacturing Technology*, vol. 73, pp. 267–277, 2014.

[3] S. Ahmed, T. Koseki, and H.-J. Kim, "Detent force reduction for a novel transverse flux permanent magnet linear synchronous motor without compromising stroke length," in *2017 11th International Symposium on Linear Drives for Industry Applications (LDIA)*. IEEE, 2017, pp. 1–5.

[4] Y. Luo, Y. Chen, H.-S. Ahn, and Y. Pi, "Fractional order robust control for cogging effect compensation in pmsm position servo systems: stability analysis and experiments," *Control Engineering Practice*, vol. 18, no. 9, pp. 1022–1036, 2010.

[5] N. Mooren, M. van Meer, G. Witvoet, and T. Oomen, "Compensating torque ripples in a coarse pointing mechanism for free-space optical communication: A gaussian process repetitive control approach," *Mechatronics*, vol. 97, p. 103107, 2024.

[6] K. K. Tan, T. H. Lee, H. Dou, and S. Zhao, "Force ripple suppression in iron-core permanent magnet linear motors using an adaptive dither," *Journal of the Franklin Institute*, vol. 341, no. 4, pp. 375–390, 2004.

[7] T. Beckers and S. Hirche, "Stability of gaussian process state space models," in *2016 European Control Conference (ECC)*. IEEE, 2016, pp. 2275–2281.

[8] H. Jung and S. Oh, "Gaussian process and disturbance observer based control for disturbance rejection," in *2022 IEEE 17th International Conference on Advanced Motion Control (AMC)*. IEEE, 2022, pp. 94–99.

[9] C. E. Rasmussen, C. K. Williams *et al.*, *Gaussian processes for machine learning*. Springer, 2006, vol. 1.


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